

Quality Assurance Engineering Intern – Assessment

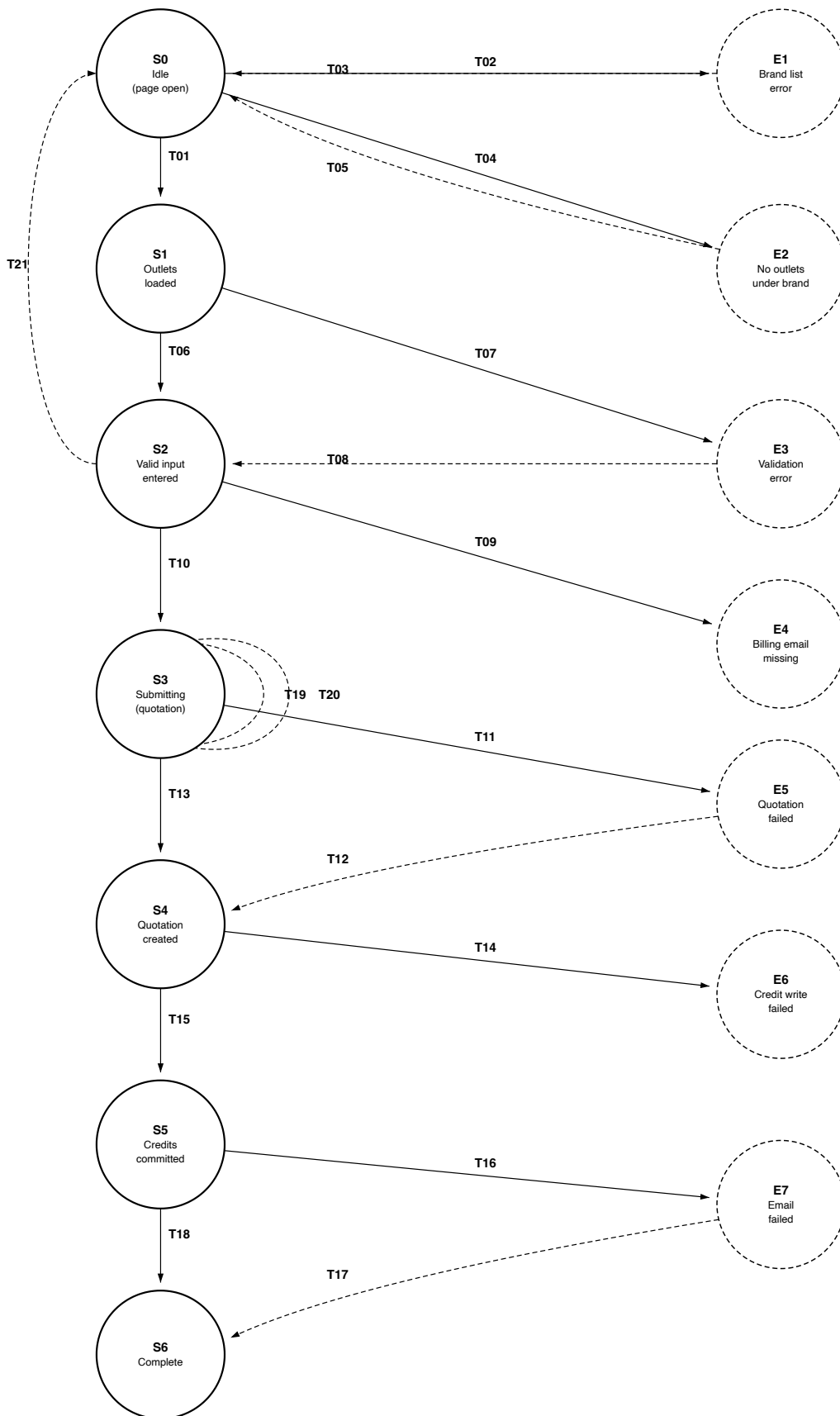
SMS Credit Reload feature (MikeTango / GoSMS) · Syed Muhammad Aziz Ghaus · azizghaus14@gmail.com · +60 17-414 7089

1. Assumptions

- SMS credits belong to a **company (billing entity)**, not to a brand. One brand can span several companies, so a single submission may bill more than one entity.
- The **quotation is the commit point**: credits are added only once QuickBooks confirms the quotation was generated.
- Sending the quotation email is a **separate, non-blocking step**. If it fails, the quotation and credits stay valid and the email is retried, because reversing credits the client has already been quoted for is the worse outcome.
- Crediting several companies happens in **one atomic transaction**, so a partial top-up cannot occur.
- Each submission carries an **idempotency key**, so a retry or double click cannot produce a second quotation or double credits.

2. State transition model

Following the four elements of state transition testing: **states** (the circles), **transitions** (the arrows), **events** (what triggers a transition) and **actions** (what the system does in response, including the message shown). Arrows are numbered; the event and action for each are listed in the transition table below.



3. Transition table

ID	From	Event	Action (system response / message)	To
T01	S0	Select a brand that has outlets	Show outlet list with company (billing entity) and current balance	S1
T02	S0	Open page, brand list fails to load	ERR-01 "Unable to load brands. Please try again." + Retry	E1

T03	E1	Retry after service restored	Brand list displayed	S0
T04	S0	Select a brand with no outlets	ERR-02 "No outlets found under this brand."	E2
T05	E2	Return to brand selection	Selection cleared	S0
T06	S1	Enter valid top-up amounts	NEXT enabled	S2
T07	S1	Enter invalid amount	ERR-03 inline message; entered values kept	E3
T08	E3	Correct the input	Error cleared, NEXT enabled	S2
T09	S2	Click NEXT with missing/invalid billing email	ERR-04 naming the company; no quotation, no credits	E4
T10	S2	Click NEXT with all data valid	Button locked, idempotency key created	S3
T11	S3	QuickBooks error or timeout	ERR-05 "No credits have been added." + Retry	E5
T12	E5	Retry with the same idempotency key	Exactly one quotation created	S4
T13	S3	Quotation generated	Quotation number issued, line item per company	S4
T14	S4	Credit transaction fails	ERR-06, transaction rolled back, quotation voided/flagged, finance alerted	E6
T15	S4	Credits committed atomically	balance = balance + top-up, per company	S5
T16	S5	Quotation email fails to send	WARN-01; credits and quotation remain valid; Resend offered	E7
T17	E7	Resend (manual or queued)	Email delivered; credits not applied again	S6
T18	S5	Quotation email sent	Audit log written; success screen shown	S6
T19	S3	Second submit (double click)	Blocked by idempotency key; no duplicate quotation or credits	S3
T20	S3	Session expires mid-submission	Re-authentication prompt; flow resumes once, single credit	S3
T21	S2	Cancel or navigate away	All input discarded; no quotation, no credits, no audit entry	S0

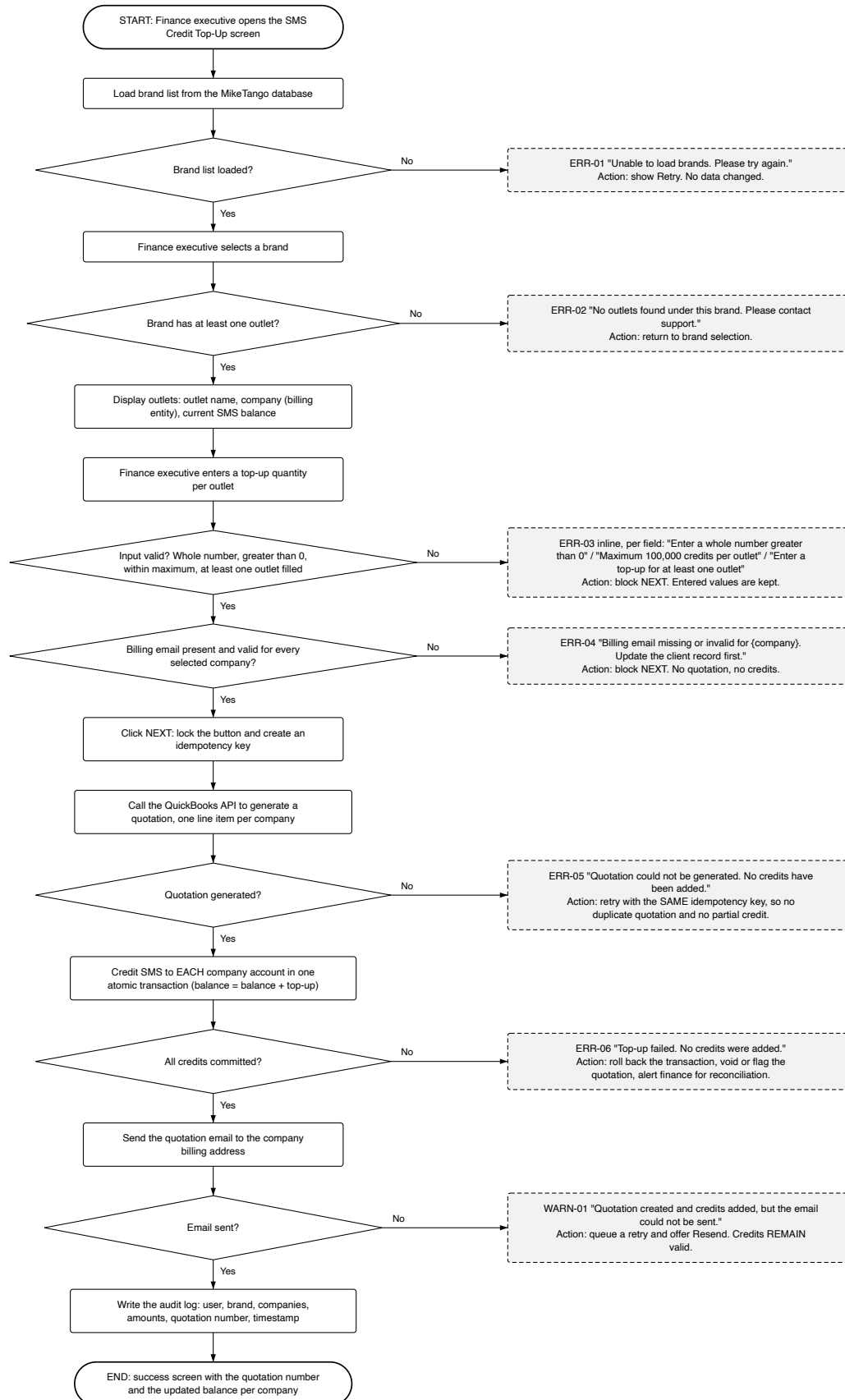
Coverage: every transition above is exercised by exactly one test case, and no transition is tested twice. T07 is a single transition triggered by six different invalid inputs, so it is written as one data-driven case (TC-07) rather than six near-identical cases.

T01→TC-01 T02→TC-02 T03→TC-03 T04→TC-04 T05→TC-05 T06→TC-06 T07→TC-07 T08→TC-08 T09→TC-09
T10→TC-10 T11→TC-11 T12→TC-12 T13→TC-13 T14→TC-14 T15→TC-15 T16→TC-16 T17→TC-17 T18→TC-18
T19→TC-19 T20→TC-20 T21→TC-21

4. Business rules (traceability)

ID	Rule
R1	SMS credits are charged and credited to the company (billing entity), never to the brand as a whole
R2	A single submission may cover outlets belonging to different companies
R3	Only outlets with a top-up entered are quoted and credited
R4	Quotation line totals, tax and grand total match the system calculation to 2 decimal places
R5	A top-up increments the existing balance rather than replacing it
R6	Every successful top-up is recorded in the audit log

5. Flowchart (including error handling and messages)



6. Test cases

Each case is traceable to a transition (T) or business rule (R), and states its own preconditions and data so it can be repeated by anyone.

ID	Ref	Title	Precondition	Steps	Expected result	Type
TC-01	T01	Select a brand that has outlets	Brand 'SilverTree' has 3 outlets across 2 companies	Select 'SilverTree'	Outlet list shows all 3 outlets with outlet name, company and current SMS balance	Positive
TC-02	T02	Brand list fails to load	Brand list service unavailable	Open the Top-Up page	ERR-01 shown with Retry. No data written anywhere	Negative
TC-03	T03	Recover from brand list failure	ERR-01 displayed; service restored	Click Retry	Brand list loads; back in S0	Recovery
TC-04	T04	Brand with no outlets	Brand 'EmptyBrand' has 0 active outlets	Select 'EmptyBrand'	ERR-02 shown; NEXT unavailable	Negative
TC-05	T05	Return from the no-outlet state	ERR-02 displayed	Choose another brand	Selection cleared and the new brand's outlets load	Recovery
TC-06	T06	Enter valid top-up quantities	Outlets displayed	Enter 500 for outlet A, 1000 for outlet B, leave C blank	Values accepted; NEXT enabled	Positive
TC-07	T07	Invalid input (data-driven, single transition)	Outlets displayed	Enter in turn: 0 / -50 / 1.5 / abc / all blank / 100001	Matching ERR-03 message each time; NEXT stays disabled; valid values preserved	Negative
TC-08	T08	Correct an invalid input	ERR-03 displayed	Replace the invalid value with 500	Error clears; NEXT enabled	Recovery
TC-09	T09	Missing billing email	Outlet B's company has no billing email	Enter valid amounts, click NEXT	ERR-04 naming the company; no quotation, no credits	Negative
TC-10	T10	Submit a valid top-up	Valid amounts; all billing emails present	Click NEXT	Button locks; idempotency key created; quotation requested	Positive
TC-11	T11	QuickBooks unavailable	QuickBooks returns error/timeout	Click NEXT	ERR-05 shown; no quotation; no credits on any company	Negative
TC-12	T12	Retry after QuickBooks failure	ERR-05 displayed; QuickBooks restored	Click Retry	Exactly one quotation exists; credits applied once	Recovery
TC-13	T13	Quotation generated	Valid submission; QuickBooks healthy	Click NEXT	Quotation created with one line item per company matching the entered amounts	Positive
TC-14	T14	Credit write fails after quotation	Quotation created; credit transaction fails mid-way	Complete submission	ERR-06; no company partially credited; quotation voided/flagged; finance alerted	Negative
TC-15	T15	Credits committed per company	Quotation created successfully	Complete submission	Each company's balance rises by its own amount; other companies unchanged	Positive
TC-16	T16	Quotation email fails	Credits committed; mail service down	Complete submission	WARN-01; credits and quotation remain valid; Resend offered	Negative
TC-17	T17	Resend the quotation email	WARN-01 displayed; mail service restored	Click Resend	Email delivered; credits not applied a second time	Recovery
TC-18	T18	Full happy path	All services healthy	Complete the flow	Email sent, audit log written, success screen shows quotation number and updated balances	Positive
TC-19	T19	Double submission	Valid amounts entered	Click NEXT twice rapidly	One quotation and one credit transaction only	Negative
TC-20	T20	Session expires during submission	Submission in flight; token expires	Complete submission	Re-authentication prompt; afterwards exactly one quotation and one set of credits	Negative
TC-21	T21	Cancel mid-flow	Amounts entered, NEXT not clicked	Navigate away, then return	No quotation, no credits, no audit entry; clean form	Negative
TC-22	R1	Credits are company-scoped	Brand spans Company X and Company Y	Top up an outlet belonging to Company X only	Only Company X is credited and billed; Company Y untouched	Business rule

TC-23	R2	Several companies in one submission	Brand spans Company X and Company Y	Top up 500 for X's outlet and 800 for Y's outlet	Charges separated per billing entity, each emailed to its own address; X +500, Y +800	Business rule
TC-24	R3	Partial fill of the outlet list	Brand has 3 outlets	Fill outlet A only	Only A's company is quoted and credited; B and C excluded entirely	Business rule
TC-25	R4	Quotation totals and rounding	Amounts producing a fractional tax value	Submit	Line totals, tax and grand total match to 2 decimal places with no rounding drift	Business rule
TC-26	R5	Balance is incremented, not overwritten	Company X holds 2,000 credits	Top up 500	Balance becomes 2,500	Business rule
TC-27	R6	Audit trail completeness	Successful top-up	Complete flow, open audit log	Entry records user, brand, companies, amounts, quotation number and timestamp	Business rule

7. Highest-risk areas

- **Money and credits out of step** (TC-11, TC-12, TC-14, TC-19, TC-20). The expensive failure is a client billed without credits, or credited twice. The idempotency key and the atomic transaction are what prevent it.
- **Crediting the wrong entity** (TC-22, TC-23, TC-24). The brand-to-company split is the subtlest rule in the feature, so it gets dedicated tests.
- **Silent partial success** (TC-14, TC-16). The user must always be told exactly which of quotation, credits and email succeeded.